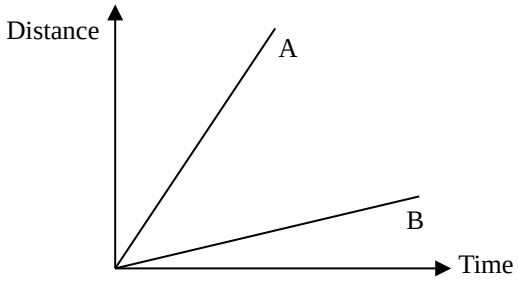
SCIENCE 1.6 **Name:** ***ANSWERS***

WORKSHEET THREE **DISTANCE-TIME GRAPHS**

1. For each of the following distance-time graphs describe the journey that the graph represents.

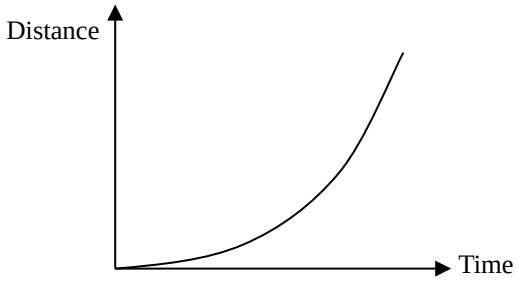
(a)



A ***high constant speed***

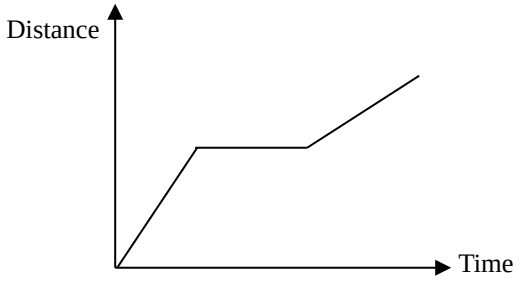
B ***low constant speed***

(b)



acceleration

(c)

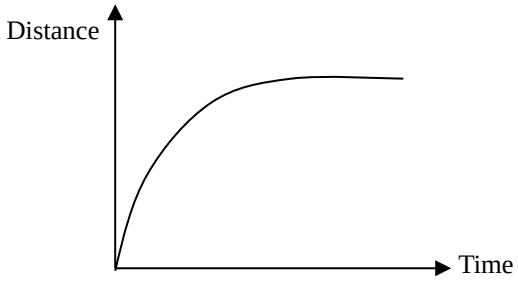


constant speed

stationary

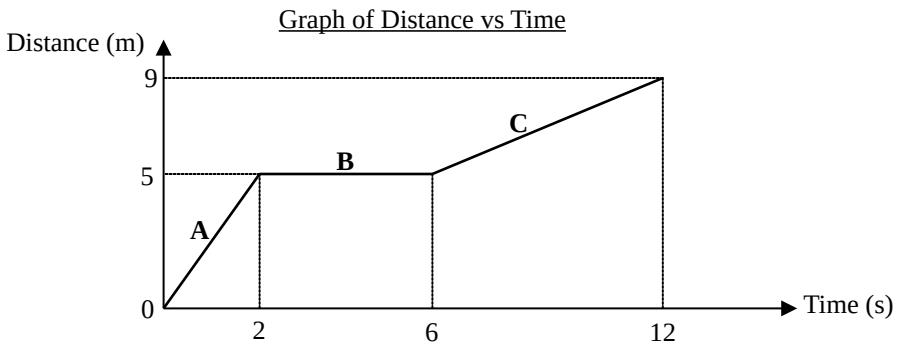
lower constant speed

(d)



deceleration to stationary

2. The motion of a car is shown on the following graph.



(a) Describe the motion in each section of the journey.

A. ***constant speed***

B. ***stationary***

C. ***slower constant speed***

(b) What distance has been travelled after 2 seconds? ***5m*** after 6 seconds? ***5m***

after 12 seconds? ***9m***

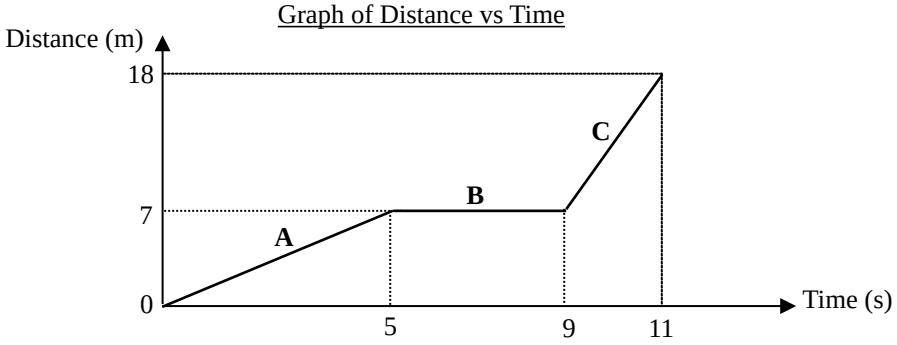
(c) Calculate the speed in

Section A $\boldsymbol{v} = \frac{rise}{run} = \frac{5}{2} = 2.5\ ms^{-1}$

Section B $\boldsymbol{v} = \frac{rise}{run} = \frac{0}{4} = 0\ ms^{-1}$

Section C $\boldsymbol{v} = \frac{rise}{run} = \frac{4}{6} = 0.67\ ms^{-1}$

3.



Calculate the speed in

Section A $\boldsymbol{v} = \frac{rise}{run} = \frac{7}{5} = 1.4\ ms^{-1}$

section B $\boldsymbol{v} = \frac{rise}{run} = \frac{0}{4} = 0\ ms^{-1}$

section C $\boldsymbol{v} = \frac{rise}{run} = \frac{11}{2} = 5.5\ ms^{-1}$